



DEPARTMENT OF
COMPUTER SCIENCE



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CURRICULAR DEVELOPMENT ACTIVITY INTERNSHIP REPORT

BACHELOR IN COMPUTER SCIENCE
APDC — COMPANY INTERNSHIP REPORT — GMV

NOVA University Lisbon
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Curricular Development Activity Internship Report

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[1] The $\LaTeX$ Project team. *$\LaTeX$ – A document preparation system*. 1985. URL: <https://www.latex-project.org>.

[2] J. M. Lourenço. *The NOVAthesis $\LaTeX$ Template User's Manual*. NOVA University Lisbon. 2021. URL: <https://github.com/joaomloureiro/novathesis/raw/main/template.pdf>.

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ACRONYMS

BeiDou	BeiDou (<i>p. 1</i>)
CDA	Curricular Development Activity (<i>p. 1</i>)
Galileo	Galileo (<i>p. 1</i>)
GLONASS	Globalnaya Navigatsionnaya Sputnikovaya Sistema (<i>p. 1</i>)
GMV	GMVIS SKYSOFT, S.A (<i>p. 1</i>)
GNSS	Global Navigation Satellite System (<i>pp. 1, 2</i>)
GPS	Global Positioning System (<i>p. 1</i>)
RS-485	Recommended Standard 485 (<i>pp. 2–4</i>)
SDR	Software-Defined Radio (<i>pp. 1–4</i>)

INTRODUCTION

This first chapter serves as a brief introduction to the internship conducted within the scope of Curricular Development Activity (CDA) at GMVIS SKYSOFT, S.A (GMV) It outlines the problems to be solved and the proposed solutions.

1.1 Context and Motivation

Global Navigation Satellite System (GNSS) are a family of satellite-based positioning systems, usually referred as a constellation, that provide geolocation and time information to a receiver anywhere on Earth. There are several different GNSS constellations in operation today, including the United States' Global Positioning System (GPS), Russia's Globalnaya Navigatsionnaya Sputnikovaya Sistema (GLONASS), the European Union's Galileo (Galileo)[2], and China's BeiDou (BeiDou). These systems consist of a network of satellites that transmit signals to receivers, which then use the signals to determine their location.

As reliance on GNSS technology grows, so does the need for accurate and reliable testing and validation of GNSS receivers and the systems that are built around them. [8]

As per [3], a signal simulator/generator is a tool used to "simulate GNSS signals that are transmitted by navigation satellites, and received by an antenna". GNSS simulators are essential for testing and validating GNSS receivers as they serve as an easy to use signal source, they enable engineers to create a controlled testing environment, thus allowing the testing of receivers under various repeatable scenarios. Signal simulators also provide the ability to simulate signals without the need of outdoor tests or long waits for satellites to be in a desired position [7], simulators may also aid in the testing new types of GNSS signals to be used in newer versions of existing systems.

Software-Defined Radio (SDR) is a radio communication paradigm in which signal components usually implemented in hardware, are instead implemented through software. This approach offers greater flexibility and adaptability, as it allows the same hardware to be modulated in different ways without requiring physical changes to the hardware itself.[4] [6] [ix-x][1] SDR platforms have been gaining popularity in recent years due to

their versatility, and, in the scope of this internship, they provide an ideal foundation for developing a flexible and efficient GNSS signal generator. [ix-x][1]

Recommended Standard 485 (RS-485) is a standard for serial communication that is commonly used in industrial applications. It allows for long-distance communication and is resistant to electrical noise, making it ideal for use in environments where electromagnetic interference is present.[5] This protocol will enable reliable data exchange between the GNSS signal generator and external devices, ensuring accurate and timely communication.

1.2 Problem

The key challenge for a signal generator is providing capacity to coherently simulate the complex signals received from multiple GNSS constellations, a generator must usually ensure: Accuracy, Precision, Stability, Range and Resolution.[3] Without these capabilities, the effectiveness of the signal generator is significantly compromised, leading to potential failures in real-world applications. Thus without the proper tooling, engineers are constrained to a manual and more costly testing process, that not only is unable or hard to support automated testing but also increases the time and effort required to validate GNSS receivers.[7] Furthermore, the almost bare-metal nature of hardware communication and low-level approach to existing API's demands careful design to ensure reliability and real-time performance.

1.3 Proposed Work

The internship aims to develop an interface that closes the gap by designing an easy-to-use C++ tool for the programmatic control of a GNSS signal generator. This interface will abstract the complexities of hardware communication and provide a more user-friendly environment for engineers to design and execute their tests. The internship can be split into three key phases:

Phase 1 SDR Control:

In this phase, the focus will be on developing the SDR control interface. This will involve creating a C++ API that allows users to interact with the GNSS signal generator at a high level, without needing to understand the low-level details of the hardware. The API will provide functions for configuring the signal generator, starting and stopping signal generation, and monitoring the status of the generator.

Phase 2 Parameter Management:

In this phase, the focus will be on developing a parameter management system that allows users to easily configure and manage the various parameters of the GNSS signal generator. This will enable users to quickly and efficiently set up their test scenarios, without having to manually adjust each parameter through low-level API calls.

Phase 3 RS-485 communication:

In this phase, the focus will be on developing the RS-485 communication interface for the GNSS signal generator. This will involve creating an API that allows users to interact with external equipment via the RS-485 interface. The API will provide functions for interface configuration, command dispatch and response handling.

1.4 Objectives

The main objectives of this internship are:

- Acquire working knowledge of the target SDR hardware architecture and its interface.
- Understand RS-485 communication protocol and its application in instrument control.
- Implement C++ modules for SDR control via the vendor API.
- Develop interfaces for parameter management and test scenario configuration.
- Implement RS-485 communication layer for exchange of commands and data with external equipment.
- Validate the overall system functionality through functional and performance testing.
- Produce technical documentation and a final report for the developed interfaces and modules.

1.5 Work Plan & Schedule

Work Plan & Schedule			
Phase	Task	Duration	Dates
Onboarding and Studying	Introduction to the signal generator architecture; study of SDR APIs; RS-485 protocol fundamentals; review of existing codebase and tooling at GMVIS Skysoft	3 weeks	Mar 11 - Mar 31
SDR Control Modules	Development of C++ API for SDR control; implementation of signal generation functions; integration with vendor API and unit testing	4 weeks	Apr 01 - Apr 28
Parameter Management	Development of parameter management system; implementation of user-friendly configuration interface; integration with SDR control modules	4 weeks	Apr 29 - May 26
RS-485 Communication	Design and Implement of an RS-485 communication layer for exchange of commands and data with external equipment	3 weeks	May 27 - Jun 16
Integration, Testing & Documentation	Integration of all components; system testing and validation; preparation of user manuals and technical documentation	5 weeks	Jun 17 - Jul 17

1.6 Conclusion

This chapter as thus introduced some of the context and motivation for the work carried out throughout the internship. The objectives and work plan were made clear, providing a roadmap for the development of the SDR control system and its integration with external equipment. The possible challenges due to the complexity of the SDR architecture and the RS-485 communication protocol were also acknowledged.

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